



Inorganic Chemistry Communications

Available online 14 May 2026, 116832

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Interface engineering with a hydrophilic and zincophilic layer enables stable zinc metal anode for aqueous zinc-ion batteries



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Highlights

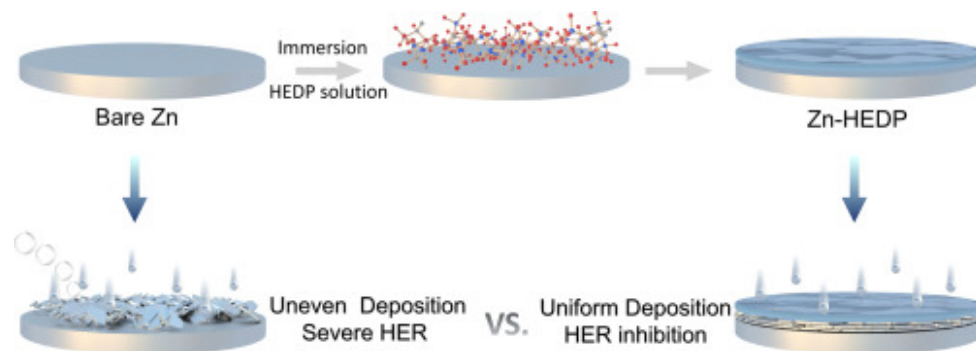
- HEDP is employed as a chelating agent to generate protective layer on Zn foil.
- The hydrophilic and zincophilic interface regulates uniform zinc deposition and inhibits side reactions.

- The Zn-HEDP electrode demonstrates an exceptional cycling stability and capacity retention.

Abstract

Hydroxyethylidene diphosphonic acid (HEDP) is utilized to fabricate an artificial functional interface on Zn foil. Owing to its hydrophilicity and zincophilicity, the as-formed protective layer effectively regulates the behavior of Zn deposition as well as enhances the durability of Zn anode. On one hand, this layer as a physical barrier prevents the direct contact between the aqueous electrolyte and Zn metal, thereby suppressing the parasitic side reactions. On the other hand, the as-fabricated Zn-HEDP electrode exhibits strong zincophilicity, ensuring the uniform Zn plating. The Zn-HEDP demonstrates an exceptional cycling stability in the symmetric cell, maintaining a stable operation for over 2000 h at a current density of $1 \text{ mA cm}^{-2}/1 \text{ mAh cm}^{-2}$. Furthermore, the Zn-HEDP//MnO₂ full cell achieves a capacity retention of ~70% after 500 cycles. This study employs the hydrophilic modifiers to engineer a functional interface on Zn foil, offering a promising strategy for developing stable zinc metal anodes.

Graphical abstract



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Keywords

Zinc-ion batteries; Zinc anode; Interface engineering; Hydrophilicity; Zincophilicity

1. Introduction

Aqueous zinc ion batteries have garnered significant attention in the field of energy storage owing to their intrinsic properties, due to the low redox potential, high theoretical specific capacity, and cost-effectiveness of zinc metal anode. Furthermore, its compatibility with aqueous electrolytes ensures inherent safety, which is a critical advantage over the other battery systems [1]. Despite these merits, the large-scale deployment of zinc anodes is severely impeded by two critical challenges including uncontrollable dendrite growth and water-induced side reactions, which cause the consumption of active materials and performance degradation of battery [2].

To enhance the performance of aqueous zinc ion batteries, interfacial engineering, electrolyte optimization, and separator modification have been investigated as effective strategies by focusing on the different components. [3], [4], [5], [6] Recently, surface modification of Zn metal with artificial interface layers has been recognized as a viable approach to address the above issues and promote the overall performance of zinc anodes [7]. Among them, hydrophobic modifiers can constrain the water-

involved side reactions and extend the stable operation of Zn anodes [8]. Yet, the zincophilicity is usually compromised by the low polarity of these modifiers, which is detrimental to uniform Zn deposition and favorable orientation [9]. As an alternative, the hydrophilic modifiers perform better in guiding Zn plating, while their dissolution or detachment in the aqueous electrolyte inevitably causes the performance degradation of device [10]. Therefore, it is urgent to develop scalable and facile strategies for constructing functional interfaces firmly anchoring to zinc surface, thereby regulating the subsequent Zn deposition [11].

Hydroxyethylidene diphosphonic acid (HEDP) is a phosphonic acid featuring an ethylene backbone and two phosphonic acid groups [12]. Its stable C—P bonds impart notable chemical properties, including strong binding ability and high chemical stability [13]. In particular, the phosphonic acid groups in HEDP molecule can form stable six-membered chelate structures adhering to metal surface, so HEDP has been directly added to the electrolyte of the battery as an additive [14], but the electrolyte additive not only increases the cost but also prevents adverse effects on ion transport kinetics in the electrolyte. Herein in this work, HEDP is employed as a chelating agent to generate a protective layer on Zn foil and pre-construct Zn-HEDP anodes. Notably, the as-fabricated stable layer resisting hydrolysis ensures the persistent adherence to zinc surface and isolates active zinc from the electrolyte [15], [16]. Meanwhile, the strong electrostatic interaction of phosphonic groups in HEDP with Zn^{2+} ions regulates the deposition of zinc metal [17]. This work provides fundamental insights into the design and construction of stable Zn anodes via interface engineering.

2. Experimental

2.1. Preparation of Zn-HEDP electrode

Commercial Zn foil (100 μm thick) was cut into circular electrodes with a diameter of 12 mm. Subsequently, the electrodes were subjected to a sequential cleaning by immersion in ethanol/deionized water, followed by drying in oven at 60 °C for 2 h. The cleaned Zn foil was immersed in an aqueous solution containing 0.1 wt% hydroxyethylidene diphosphonic acid (HEDP) for different reaction durations of 0.5 h, 1 h, and 3 h, respectively. Afterward, the electrodes were rinsed with deionized water for 3–5 times and then dried in a vacuum oven at 60 °C overnight, yielding the electrodes of Zn-HEDP-0.5, Zn-HEDP-1, and Zn-HEDP-3, respectively.

2.2. Material characterizations

The morphology and microstructure of electrodes were characterized using a field-emission scanning electron microscope (FE-SEM, Hitachi SU8600) operated at 10 kV. X-ray diffraction (XRD, Shimadzu XRD-6000) with Cu- K_{α} radiation was used to analyze the phase composition. Fourier-transform infrared spectroscopy (FT-IR, Nicolet IS10, Thermo-Fisher Scientific) was conducted in the wavenumber range of 4000–400 cm^{-1} . Raman spectroscopy was performed using a confocal micro-Raman spectrometer (DXR3xi, Thermo-Fisher Scientific) with an excitation wavelength of 532 nm. The chemical states of elements were analyzed using X-ray photoelectron spectroscopy (XPS, ESCA-LAB250-Xi, Thermo-Fisher Scientific). The electrolyte wettability onto the electrode was measured through contact angle measurements with a contact angle goniometer (SINDIN, China). An optical microscope (Smartzoom 5, Zeiss) integrated with a simulated in-situ electrochemical cell was utilized to monitor real-time dynamic observations during the Zn deposition.

2.3. Electrochemical measurements

The electrochemical performance evaluations were conducted using CR2032-type coin cells assembled in air. Symmetric cells were fabricated with bare Zn and Zn-HEDP as the electrodes, separated by a glass fiber separator (Whatman GF/D) and filled with 100 μL of 2 M ZnSO_4 aqueous solution as the electrolyte. For half-cell configurations, Cu foil serves as the counter electrode while bare Zn or Zn-HEDP was used as the working electrode. Full cells were assembled using α - MnO_2 cathode paired with bare Zn or Zn-HEDP as the anode, with the electrolyte consisting of 100 μL of 2 M ZnSO_4 solution supplemented with 0.1 M MnSO_4 . The active material loading on the cathode in this study was approximately $\sim 1.5 \text{ mg cm}^{-2}$.

Galvanostatic charge-discharge measurements were performed with a LAND CT2001A battery tester at 25 °C under multiple current densities. Electrochemical impedance spectroscopy (EIS), linear sweep voltammetry (LSV), cyclic voltammetry (CV), chronoamperometry (CA), and Tafel polarization measurements were performed on a CHI-760E electrochemical workstation. For EIS analyses, the frequency range was set from 0.01 Hz to 10^5 Hz with an AC amplitude of 5 mV. LSV tests were measured in a three-electrode configuration, comprising a saturated calomel electrode (SCE) as the reference electrode, platinum sheet as the counter electrode, and Zn-HEDP or bare Zn as the working electrode, with a scan rate of 5 mV s^{-1} and a potential range of -1.1 to -2.1 V (vs. SCE). CV measurements were conducted at 0.2 mV s^{-1} over a potential window of 0.6 – 1.8 V (vs. SCE). CA tests were performed on symmetric cells with an initial applied potential of -150 mV . Tafel polarization analysis utilized the same three-electrode configuration with a potential range of -1.3 – 0.7 V (vs. SCE) and a scan rate of 1 mV s^{-1} , where the corrosion current densities were determined through Tafel extrapolation.

The utilization efficiency of the zinc anode was calculated using the depth of discharge (DOD) formula as follows:

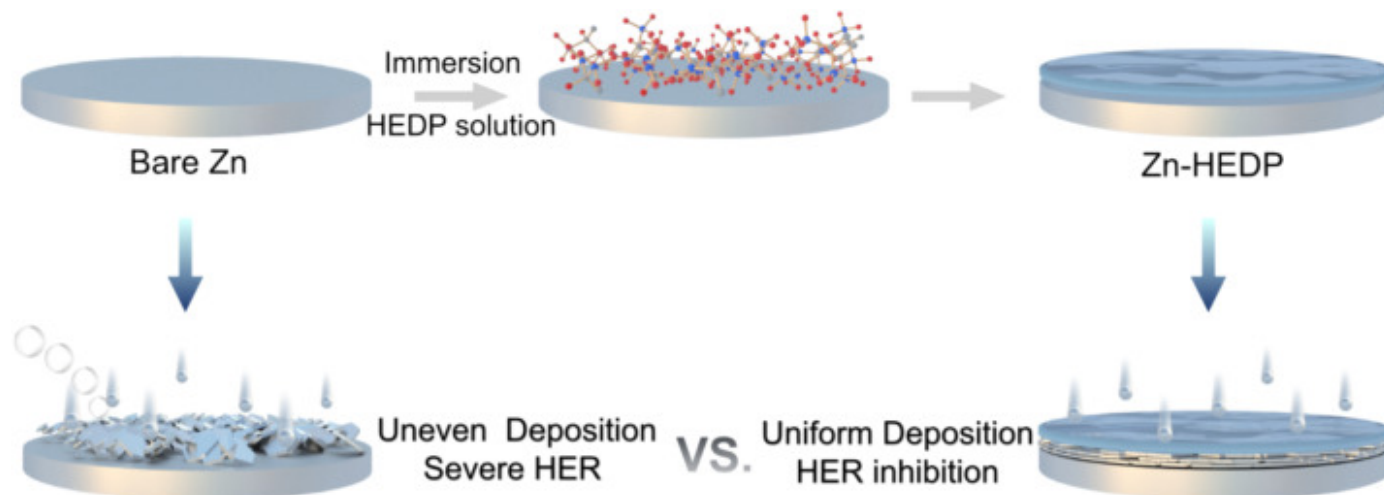
$$\text{DOD}(\%) = \left(\frac{y}{C_{\text{Zn,Volume}} \times x \times 10^{-4}} \right) \times 100\% = \frac{y}{0.585x} \times 100\% \quad (1)$$

In Eq. (1), x (μm) is the thickness of the zinc foil, y (mAh cm^{-2}) is the actual zinc capacity consumed during the electrochemical test, and $C_{\text{Zn,Volume}}$ is the theoretical volumetric capacity (5854 mAh cm^{-3}). Therefore, according to the areal capacities set in the experiments, the single-cycle utilization efficiencies of the zinc anode under different test conditions are as follows: 1.7% at $1 \text{ mA cm}^{-2}/1 \text{ mAh cm}^{-2}$, 3.4% at $2 \text{ mA cm}^{-2}/2 \text{ mAh cm}^{-2}$, and 8.5% under the high-rate/high-capacity condition of $5 \text{ mA cm}^{-2}/5 \text{ mAh cm}^{-2}$.

3. Results and discussion

3.1. Fabrication and characterizations of Zn-HEDP electrode

Fig. 1 illustrates the schematic fabrication and Zn deposition process of the Zn-HEDP electrode. The adsorption of HEDP molecules on Zn foil is primarily achieved through chelation reactions between phosphonic acid groups and hydroxyl oxygen atoms in HEDP and metal ions [18]. Under neutral conditions, two phosphonic acid groups ($-\text{PO}_3\text{H}_2$) in each HEDP molecule undergo a partial deprotonation to form anionic $-\text{O}^-$ groups, which subsequently coordinate with Zn^{2+} ions to form a stable six-membered chelating structure. Additionally, the hydroxyl oxygen atoms can participate in the secondary coordination interaction, further stabilizing the resultant chelate complex [19]. The overall coordination process can be described by the reaction equation of $\text{HEDP} + \text{Zn}^{2+} \rightarrow \text{Zn}(\text{HEDP})_x + y\text{H}^+$, where x and y represent the stoichiometric number of HEDP involved in coordination and the protons released during the coordination, respectively, as determined by the deprotonation degree of phosphonic acid groups.



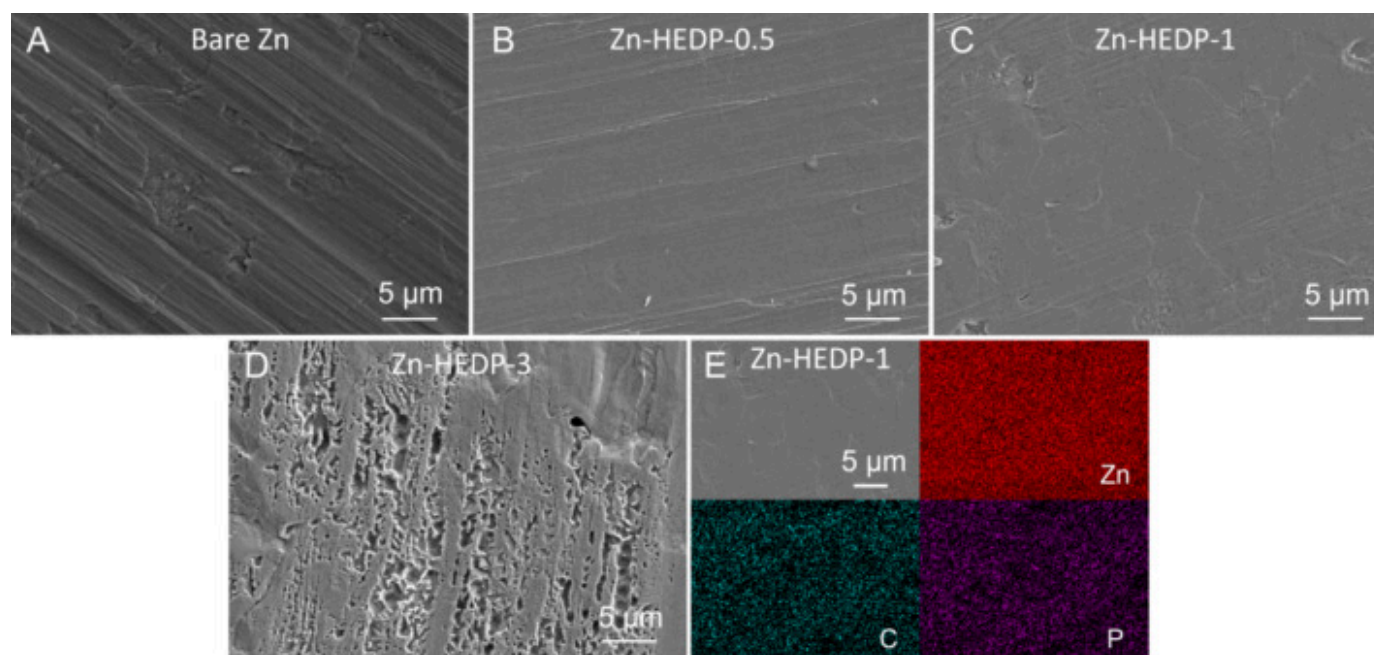
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Fig. 1. Schematic illustration of fabrication and Zn deposition of the Zn-HEDP electrode.

The microscopic morphology and structural evolution of Zn foil before and after reaction in HEDP solution were characterized by SEM. As illustrated in Fig. 2A, the as-received Zn foil (without HEDP treatment) exhibits distinct mechanical scratches. Then, Zn foil is immersed in the solution of HEDP for varying durations, and their post-immersion surface morphology is further analyzed. As shown in Fig. 2B, the scratches on Zn foil become slightly shallower after 0.5 h of immersion (Zn-HEDP-0.5), with no significant morphological alterations or coating structures. After 1 h of immersion (Zn-HEDP-1, Fig. 2C), the scratches on Zn foil are almost completely eliminated, resulting in a smooth surface. This observation may be attributed to the mild acidity and strong chelating ability of HEDP molecules [20], which continuously reacts with surface zinc. As depicted in Fig. 2D, after 3 h of immersion (Zn-HEDP-3), the flat electrode undergoes a severe surface corrosion, leading to the formation of micron-sized pore structures with non-uniform distribution. The existence of these pores would induce spatially non-uniform characteristics for the kinetics of Zn deposition/dissolution during the electrochemical cycles [21]. Considering the above SEM observations, the optimal reaction duration for Zn immersion in HEDP solution is determined to be 1 h. As depicted in Fig. 2E, elemental mapping analysis confirms the presence of Zn, C, and P on the surface of Zn foil. The P species originate from the phosphonic acid groups of HEDP, thereby validating the anchoring of HEDP onto Zn foil. Furthermore, the uniform distribution of Zn, C, and P elements across the

electrode surface indicates the formation of a homogeneous $\text{Zn}(\text{HEDP})_x$ layer, with no evidence of significant phase separation or aggregation.



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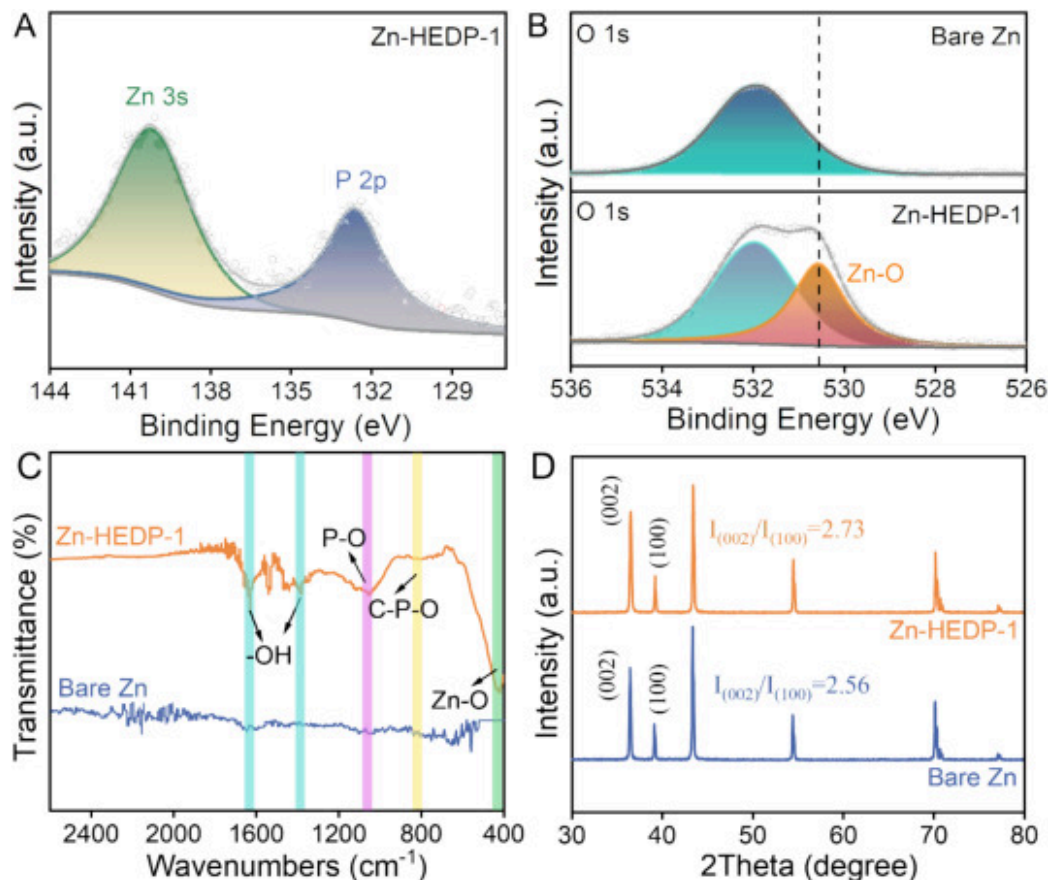
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Fig. 2. SEM images of (A) bare Zn, (B) Zn-HEDP-0.5, (C) Zn-HEDP-1, and (D) Zn-HEDP-3 electrodes; (E) EDS elemental mapping distribution of the Zn-HEDP-1 electrode.

To better illustrate the effect of treatment time with HEDP on the coating, additional EDS measurements were performed on the zinc surface under the three preparation conditions (Fig. S1). It can be observed that the P signal gradually increases, with the atomic ratio of P rising from 0.43% to 0.88% and then to 1.15%. This proves that a longer HEDP treatment time results in a thicker modification layer on the zinc surface.

To elucidate the regulation role of $\text{Zn}(\text{HEDP})_x$ interface layer, comparative analyses were performed for both Zn-HEDP-1 and bare Zn electrodes. The high-resolution P 2p XPS spectrum (Fig. 3A) exhibits two distinct peaks at 132.6 eV and 140.2 eV, which correspond to the P 2p orbitals of P—O bonds and Zn 3s orbitals, respectively [22]. This indicates that HEDP molecules form

chemical bonds with zinc metal through phosphonate groups. As shown in the O 1s spectrum (Fig. 3B), the Zn-HEDP-1 exhibits a deconvolution peak at 530.6 eV that is absent in the spectrum of bare Zn. This new peak is attributed to the lattice oxygen of Zn—O bonds formed during the HEDP modification [23], providing further evidence for the anchoring of HEDP onto Zn foil via coordination bonds.



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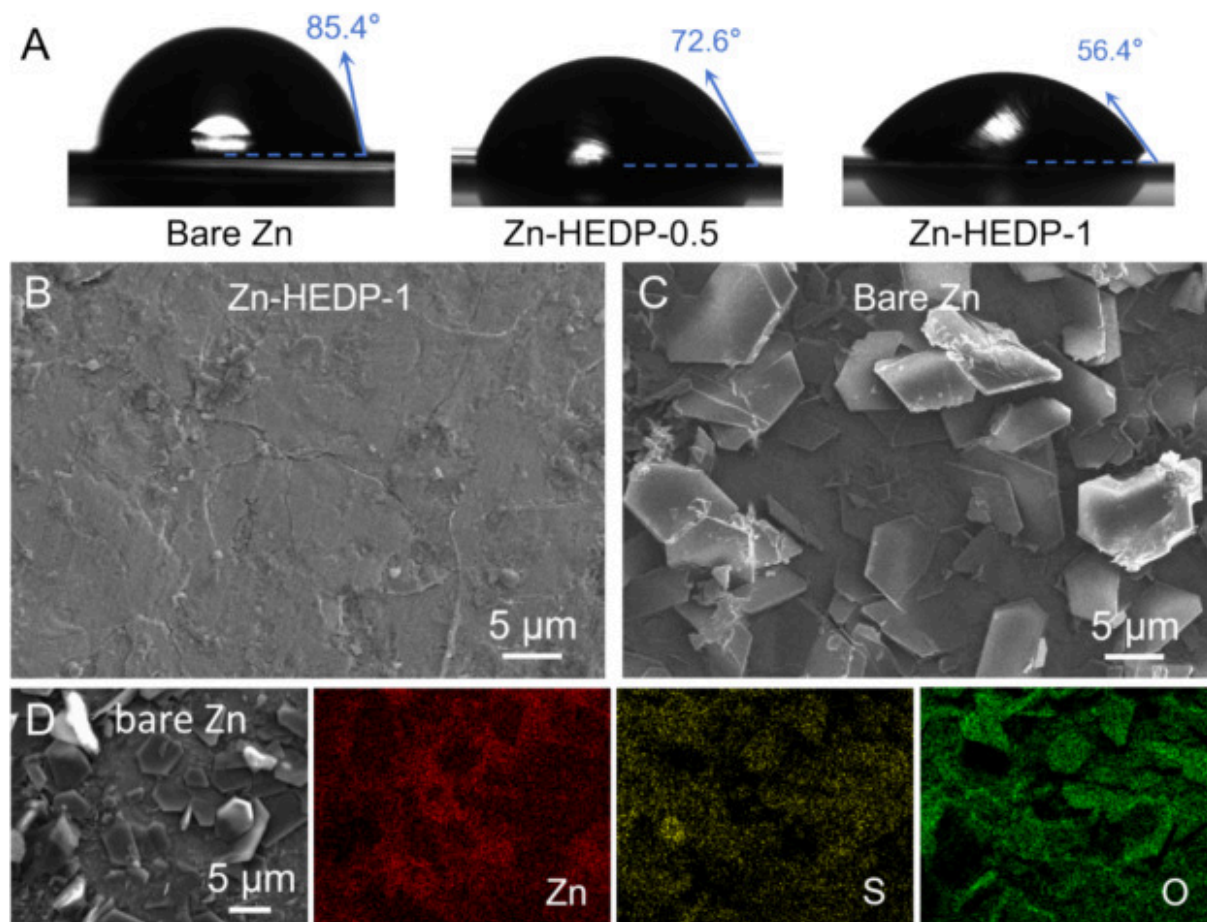
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Fig. 3. (A) Zn 3s/P 2p XPS spectrum, (B) O1s XPS spectrum, (C) FT-IR spectra, and (D) XRD patterns of Zn-HEDP-1 and bare Zn.

The FT-IR analyses (Fig. 3C) of Zn-HEDP-1 reveal a characteristic absorption peak at 419 cm⁻¹, which is related to the stretching vibration mode of Zn—O bonds [24]. This observation confirms a stable coordination interaction between Zn²⁺ ions and HEDP

molecules. The appearance of C—P—O characteristic peak at 827 cm^{-1} indicates the covalent bonding state of phosphonate groups with the organic backbone of HEDP, while the P—O asymmetric vibration at 1050 cm^{-1} validates the chemical anchoring of phosphonate groups within $\text{Zn}(\text{HEDP})_x$ layer [25]. Notably, the broadened absorption peaks at 1390 cm^{-1} and 1630 cm^{-1} are attributed to the bending vibration of hydroxyl groups and the O—H stretching vibration of adsorbed water, respectively [26]. This suggests that the $\text{Zn}(\text{HEDP})_x$ interface exhibits a good hydrophilicity, which affects the structure of Zn deposition layer as well as the reversibility of electrochemical cycles by regulating the deposition kinetics. As illustrated in Fig. 3D, no distinct differences of both Zn-HEDP and Zn electrodes in the diffraction patterns are attributed to the nanoscale thickness of $\text{Zn}(\text{HEDP})_x$ layer, which is difficult to be detected by XRD. The XRD results also show that the intensity ratio of the (002) to (100) planes on the bare Zn surface is 2.56. After HEDP treatment for one hour, the (002)/(100) intensity ratio of Zn-HEDP-1 becomes 2.73, representing an approximate change of 6%. The discrepancy hints a minimal probability that the HEDP might affect the crystallographic modification.

Dynamic contact angle measurements were employed to assess the electrode wettability (Fig. 4A). The contact angles of aqueous zinc sulfate electrolyte droplet on bare Zn, Zn-HEDP-0.5, and Zn-HEDP-1 electrodes are determined to be 85.4° , 72.6° , and 56.4° , respectively. Apparently, the Zn-HEDP-1 electrode exhibits an enhanced hydrophilicity compared to those of bare Zn and Zn-HEDP-0.5, confirming that 1 h represents the optimal modification duration for achieving a superior wettability. The hydrophilic layer facilitates the ion diffusion and uniform Zn deposition during the cycles, which is crucial for promoting the electrochemical performance of Zn anode [27].



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Fig. 4. (A) Contact angles of bare Zn, Zn-HEDP-0.5, and Zn-HEDP-1 electrodes; SEM images of (B) Zn-HEDP-1 and (C) bare Zn after immersion in 2 M ZnSO_4 solution for one week; (D) EDS elemental mapping distribution of bare Zn electrode after one-week immersion in 2 M ZnSO_4 solution.

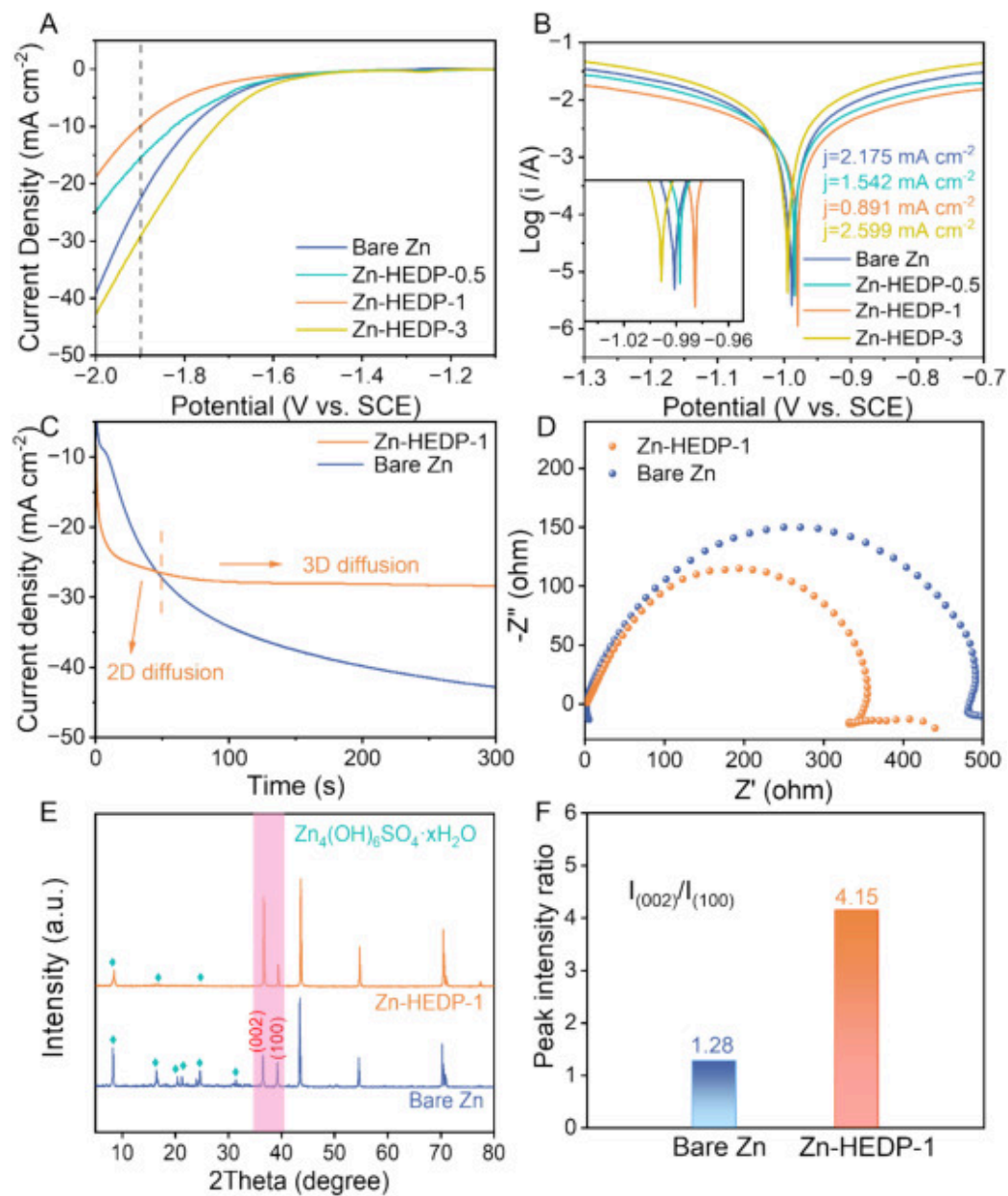
To investigate the interfacial stability, both Zn-HEDP-1 and bare Zn electrodes were respectively immersed in 2 M ZnSO_4 electrolyte for one week, followed by a thorough rinsing with deionized water. Despite the high polarity and hydrophilicity of $\text{Zn}(\text{HEDP})_x$ layer, the water-related side reactions were not induced. As shown in SEM images in Fig. 4B, the Zn-HEDP-1 electrode

maintains a slightly smooth and intact interface without observable cracks or corrosion byproducts. However, the bare Zn electrode exhibits a significant accumulation of hexagonal flake-like deposits (Fig. 4C), typical characteristic of zinc crystals. This suggests that bare Zn undergoes a dynamic equilibrium of dissolution/deposition in the electrolyte [28]. The presence of substantial deposits on bare Zn reveals that the exchange between Zn foil and zinc ions in the electrolyte leads to an interfacial degradation of the electrode even in the absence of applied voltage. In contrast, the ability of Zn-HEDP-1 to maintain a smooth surface can be attributed to the effective barrier formed via $\text{Zn}(\text{HEDP})_x$ layer, which prevents the direct contact of electrolyte and zinc metal [29]. This demonstrates the pivotal role of $\text{Zn}(\text{HEDP})_x$ interface in suppressing the corrosion and enhancing structural stability of electrode.

The attachment on Zn foil through repeated dissolution-deposition cycles in the electrolyte was further characterized by elemental mapping distribution. Fig. 4D, confirms the existence of Zn, S, and O elements in the attachment on bare Zn electrode, and the detection of S indicates the incorporation of electrolyte-derived components into the byproducts. The spatial correlation between S and O distributions as well as the layered morphology suggest that the surface of hexagonal zinc crystals has undergone an electrochemically conversion to form the electrochemically inert products, specifically $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot x\text{H}_2\text{O}$, rather than simple oxides [30].

3.2. Inhibition of side reactions and regulation of zinc deposition on Zn-HEDP

Linear sweep voltammetry (LSV) and Tafel polarization profiles demonstrate that the $\text{Zn}(\text{HEDP})_x$ interface has dramatically promoted the electrochemical performances. As presented in Fig. 5A, the current density associated with HER for Zn-HEDP-1 in 1 M Na_2SO_4 electrolyte is substantially lower than that of bare Zn within the potential range of -1.0 to -2.0 V (vs. SCE), confirming an effective suppression of direct water contact with Zn surface and kinetic inhibition of HER side reactions. The Tafel curves in Fig. 5B quantify this protective influence, revealing a positive shift in the corrosion potential from -0.990 V (bare Zn) to -0.978 V (Zn-HEDP-1) and a reduction in the corrosion current density from 2.175 to 0.891 mA cm^{-2} . This indicates a decrease in both the thermodynamic corrosion tendency and corrosion rate of Zn electrode due to the protective layer formed by the HEDP modification [31].



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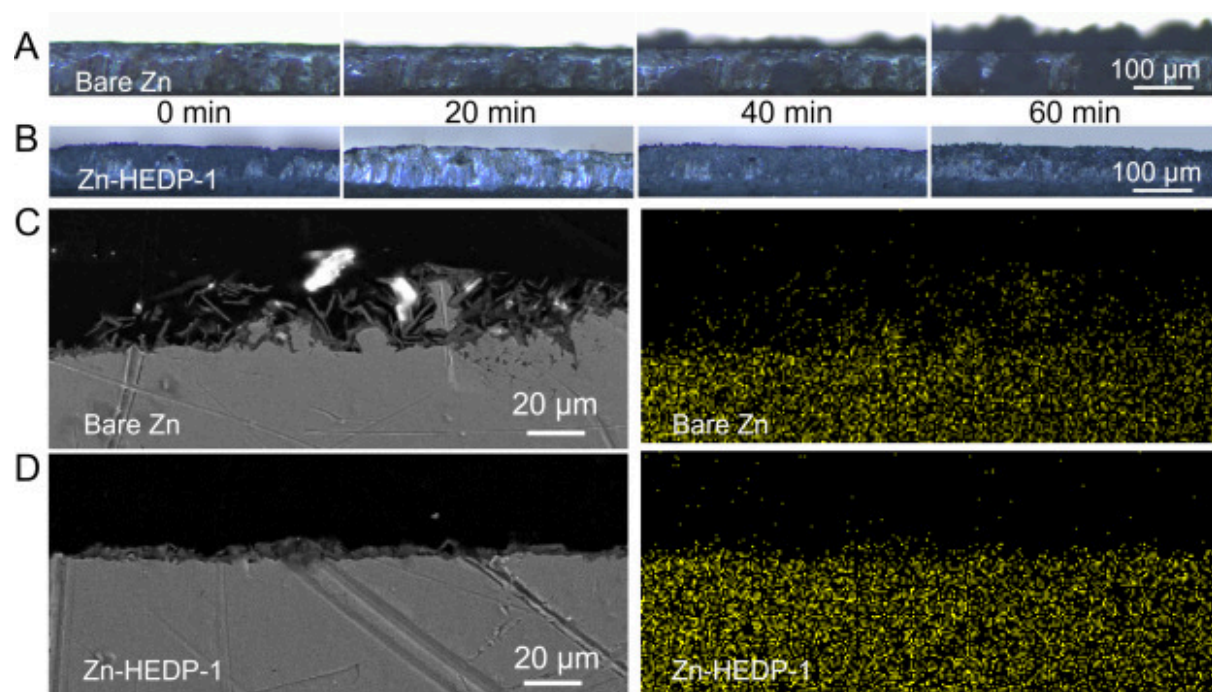
Fig. 5. (A) LSV profiles and (B) Tafel curves of Zn-HEDP-1 and bare Zn electrodes in 1 M Na₂SO₄ aqueous electrolyte at a scan rate of 5 mV s⁻¹; (C) CA curves of Zn-HEDP-1 and bare Zn electrodes at an overpotential of -150 mV; (D) Nyquist plots of symmetric cells with Zn-HEDP-1 and bare Zn electrodes at initial states; (E) XRD patterns of Zn-HEDP-1 and bare Zn after cycling for 100 h at 1 mA cm⁻²; (F) $I_{(002)}/I_{(100)}$ ratio of Zn-HEDP-1 and bare Zn.

Chronoamperometry (CA) was utilized to investigate the diffusion behaviors during zinc nucleation. During zinc deposition, dendrite formation is closely linked to the 2D diffusion mode of zinc on the surface. Therefore, a rapid transition of the zinc diffusion mode from 2D to 3D is generally beneficial for forming a flat deposition layer and suppressing dendrite formation. In the CA curve, a current approaching a plateau typically indicates a transition from the 2D diffusion mode to the 3D diffusion mode [32]. The CA curves elucidate the regulatory role of Zn(HEDP)_x interface on the deposition kinetics (Fig. 5C). Under a polarization voltage of -150 mV, the current density of bare Zn continues to rise throughout the measurement in 300 s, indicating an uncontrolled deposition/dissolution accompanied by parasitic side reactions, which leads to a continuous degradation of the electrode/electrolyte interface [33]. For the Zn-HEDP-1 electrode, a stable current plateau is achieved within 50 s, signifying the establishment of uniform deposition kinetics, and reflecting that the Zn(HEDP)_x interface promotes the completion of lateral two-dimensional (2D) diffusion of Zn²⁺ ions at the interface, and facilitates a transition into 3D diffusion for homogeneous deposition [34]. These results demonstrate that the Zn(HEDP)_x layer provides a critical kinetic control for enhancing the cycling reversibility of zinc electrode. As illustrated in Fig. 5D, the charge transfer resistance (R_{ct}) of the newly-assembled symmetric battery with Zn-HEDP-1 is reduced, indicating that the interfacial charge transfer resistance after HEDP modification is smaller than that of bare Zn, accompanied by an accelerated deposition/dissolution kinetics of zinc ions [35].

To investigate the growth mechanism of zinc dendrites during the cycles, SEM characterizations were performed to examine the morphological evolution of both Zn-HEDP-1 and bare Zn electrodes throughout repeated charge-discharge cycles. After 100 cycles at a current density of 1 mA cm⁻², the surface of bare Zn electrode is covered with vertically stacked flake-like structures and numerous vertically aligned hexagonal zinc dendrites (Fig. S2), whereas the surface of Zn-HEDP-1 remains relatively flat (Fig. S3). The additional EDS data indicate that the S and O signals on the Zn-HEDP-1 surface are very weak, suggesting that only a small amount of the byproduct basic zinc sulfate is present. In contrast, the S and O signals on the bare Zn surface are significantly stronger, indicating a higher content of byproducts. This confirms that the regulatory effect of hydrophilic layer on Zn deposition orientation and its inhibitory role in suppressing the dendrite growth [36].

Comparative XRD analyses elucidate the regulatory effect of $\text{Zn}(\text{HEDP})_x$ layer on the deposition orientation. As shown in Fig. 5E, the XRD patterns of bare Zn and Zn-HEDP-1 are compared after 100 cycles of charge-discharge at 1 mA cm^{-2} . Phase structure analysis reveals that bare Zn electrode exhibits stronger diffraction peaks of $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot x\text{H}_2\text{O}$ than those of Zn-HEDP-1, indicating more severe side reactions occurred during cycling. Quantitative analysis of (002) planes' preferred orientation (Fig. 5F) demonstrates that the $I_{(002)}/I_{(100)}$ intensity ratio for bare Zn electrode is only 1.28, reflecting a polycrystalline disordered growth, which is a characteristic closely associated with random dendrite formation and surface corrosion [37]. For the Zn-HEDP-1 electrode, a significantly increased ratio of 4.15 confirms that the interfacial layer induces a preferential orientation of Zn deposition along the (002) planes, thereby forming a highly ordered crystalline structure. This preferred orientation mechanism can be attributed to the influence of $\text{Zn}(\text{HEDP})_x$ layer on zinc-ion diffusion as well as the reduction of nucleation overpotential, ultimately facilitating the deposition along (002) planes and promoting the cycling reversibility [38].

To validate that the Zn(002)-orientation is more favorable for forming a uniform and dense Zn deposition layer on the Zn-HEDP-1 electrode, the in-situ optical microscopy was employed to dynamically monitor the deposition process in real time, as demonstrated in Fig. 6A and Fig. 6B. Under a high current density of 5 mA cm^{-2} , after 20 min of electrodeposition, the side reactions emerge on bare Zn, whereas the Zn-HEDP-1 electrode preserves a notably planar deposition. As the deposition duration extends to 40 min, the corrosion on bare Zn intensifies substantially. After 60 min, the bare Zn electrode is almost entirely covered by corrosion products. In contrast, the Zn-HEDP-1 electrode still displays a highly homogeneous zinc-deposition layer. Cross-sectional SEM images and the corresponding Zn distribution were also examined for bare Zn and Zn-HEDP electrodes cycled for 20 cycles at $2 \text{ mA cm}^{-2}/1 \text{ mAh cm}^{-2}$. The surface of bare Zn exhibits a porous structure with randomly oriented zinc flakes (Fig. 6C). This porous morphology indicates a density reduction of surface zinc layer after cycling, which may lead to the voltage polarization and loss of active zinc. In comparison, although the zinc layer on Zn-HEDP also becomes less dense than the bulk after cycling, its distribution is thinner and more uniform compared to bare Zn (Fig. 6D). This suggests an improved performance retention relative to bare Zn. The above results elucidate the optimization mechanism of interfacial layers on regulating Zn deposition process [39].



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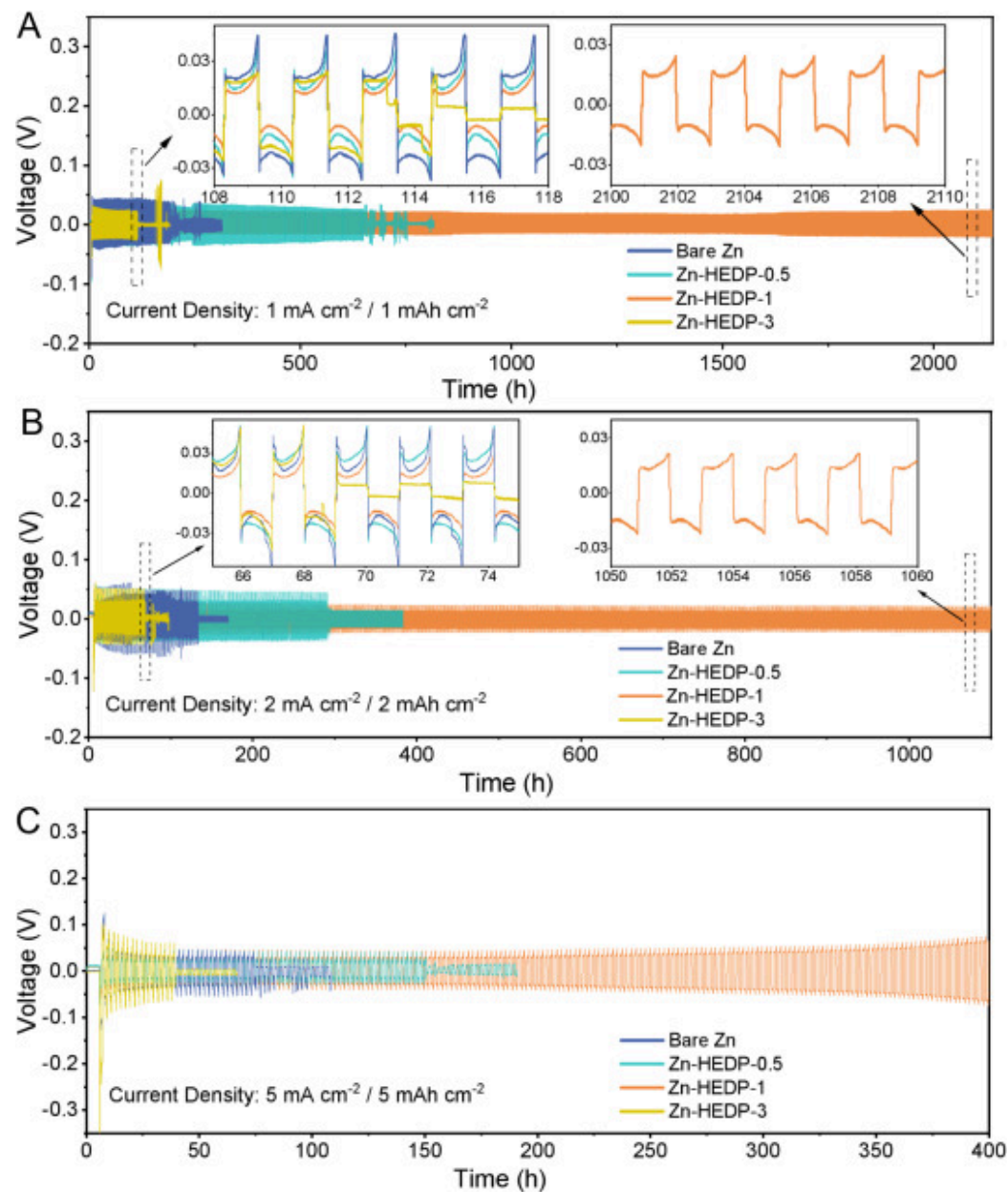
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Fig. 6. In-situ optical microscope observations of Zn plating on (A) bare Zn and (B) Zn-HEDP-1 electrodes at 5 mA cm^{-2} ; Cross-sectional SEM images and the corresponding Zn distribution of (C) bare Zn and (D) Zn-HEDP.

3.3. Electrochemical performances of Zn-HEDP

To evaluate the impact of interface layer on the electrochemical performance in the cells, symmetric cells were firstly assembled with bare Zn and Zn-HEDP electrodes. The cycling performance comparison reveals that the immersion time of Zn foil in HEDP solution determines the electrochemical stability. Under the condition of $1 \text{ mA cm}^{-2}/1 \text{ mAh cm}^{-2}$, the symmetric cell with Zn-HEDP-1 demonstrates an exceptionally long and stable cycling life exceeding 2140 h, with a polarization voltage of only 15.8 mV after 2000 cycles (Fig. 7A). In stark contrast, the bare Zn electrode fails after only 212 h, exhibiting the highest polarization voltage (110 mV) at 110 h. The symmetric cell with Zn-HEDP-0.5 shows a cycling lifetime of 648 h, whereas the Zn-HEDP-3-based symmetric cell experienced a short-circuiting after merely 108 h, although its polarization voltage remains lower than that of

bare Zn. This phenomenon highlights the critical role of reaction time in HEDP solution in determining the quality of modification layer. An optimal immersion duration of 1 h promotes the formation of a homogeneous and smooth surface, along with a stable and intact interface. In comparison, an insufficient treatment time (0.5 h) results in an incomplete coverage of modification layer, while an excessive treatment (3 h) induces the surface corrosion and pore formation, leading to a localized charge concentration and structural stress mismatch [40].



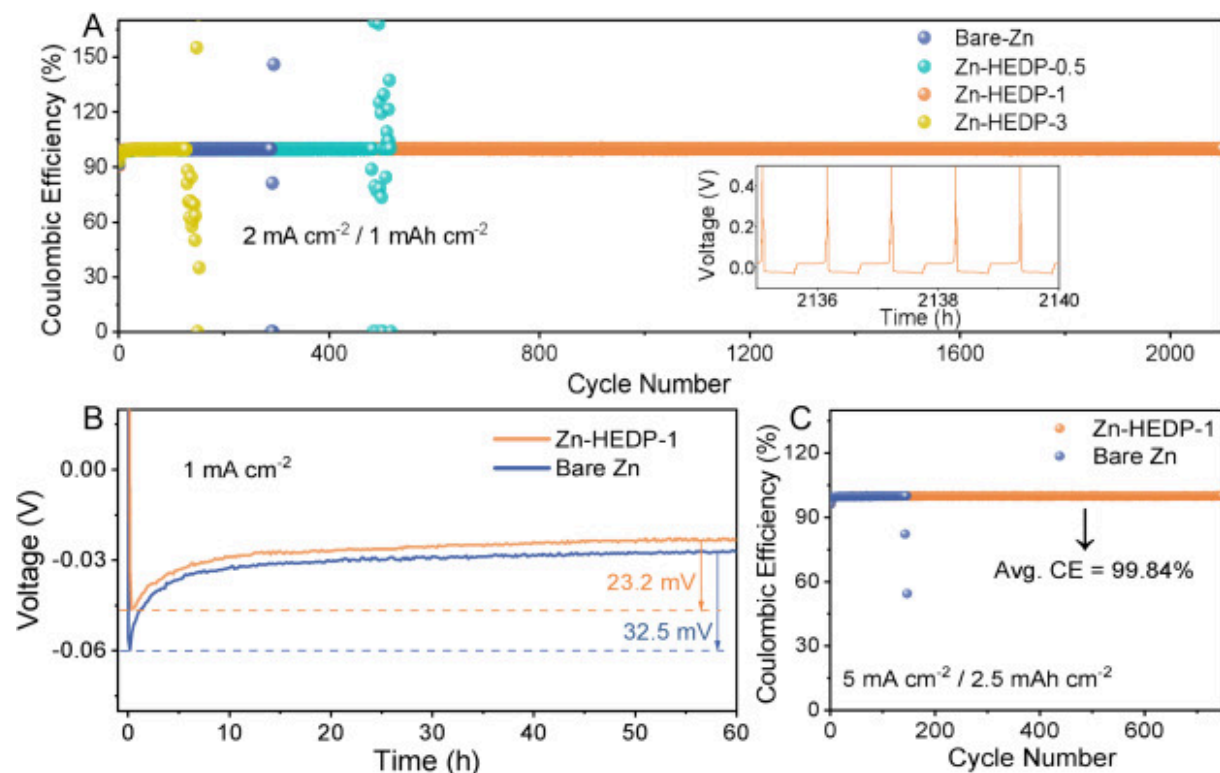
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Fig. 7. Cycling performance of symmetric cells with Zn-HEDP-1 and bare Zn electrodes at (A) $1 \text{ mA cm}^{-2}/1 \text{ mAh cm}^{-2}$, (B) $2 \text{ mA cm}^{-2}/2 \text{ mAh cm}^{-2}$, and (C) $5 \text{ mA cm}^{-2}/5 \text{ mAh cm}^{-2}$.

Parallel tests under the condition of $2 \text{ mA cm}^{-2}/2 \text{ mAh cm}^{-2}$ (Fig. 7B) demonstrate that the Zn-HEDP-1 electrode exhibits a superior cycling stability, maintaining an operational integrity for over 1100 h with a polarization voltage of merely 16.7 mV after 1000 cycles. In comparison, the bare Zn cell fails after just 140 cycles of charge-discharge, underscoring the efficacy of $\text{Zn}(\text{HEDP})_x$ interface in improving the electrode stability. Notably, the symmetric cell with Zn-HEDP-0.5 ceases the operation after 290 cycles, compared to only 69 cycles for the Zn-HEDP-3 cell. To evaluate the cycling reversibility and interface stability of the Zn-HEDP-1 electrode under harsh operating conditions, the galvanostatic cycling tests at higher current densities were also investigated. Under the condition of $5 \text{ mA cm}^{-2}/5 \text{ mAh cm}^{-2}$ (Fig. 7C), the symmetric cell assembled with bare Zn electrode experiences a short circuit after only 74 h of operation, accompanied by an initial polarization voltage difference exceeding 300 mV. This phenomenon is attributed to the exacerbated HER, severe corrosion, and excessive dendrite growth at the interface [41]. Although the Zn-HEDP-0.5 electrode reduces the voltage hysteresis to 41 mV, its performance degrades after 149 h. The Zn-HEDP-3 symmetric cell short-circuits after just 31 h, while the Zn-HEDP-1 cell maintains highly reversible zinc deposition/stripping processes under the same conditions, extending its cycle life to over 400 h. The symmetric cells were also assembled with $20 \mu\text{m}$ thick Zn foil and tested at $5 \text{ mA cm}^{-2}/2 \text{ mAh cm}^{-2}$ (Fig. S4), which show similar tendency and prove the effectiveness of HEDP modification.

To further validate the effect of interface layer on the electrochemical behavior, the half-cells of Zn//Cu and Zn-HEDP//Cu were assembled to evaluate the reaction reversibility via Coulombic efficiency (CE). Under a galvanostatic cycling at $2 \text{ mA cm}^{-2}/1 \text{ mAh cm}^{-2}$ (Fig. 8A), the half cell with Zn-HEDP-1 maintains an average CE of 99.78% over 2100 cycles, with the voltage hysteresis curve remaining stable even after 2000 cycles. In comparison, the half-cell with bare Zn exhibits a severe CE fluctuation and complete failure after only 290 cycles. Under the identical condition, the Zn-HEDP-0.5 half-cell sustains 482 cycles before failure, representing an improvement over bare Zn but remaining insufficient due to the incomplete coverage of interface layer. Notably, the Zn-HEDP-3 half-cell shows a pronounced CE fluctuation after merely 128 cycles, performing worse than bare Zn. This deterioration is attributed to the surface corrosion and structural imbalance arising from an excessive immersion, which induces the formation of “dead zinc” regions that severely hinder the process of zinc deposition/stripping, thereby degrading the CE and cycling stability [42].



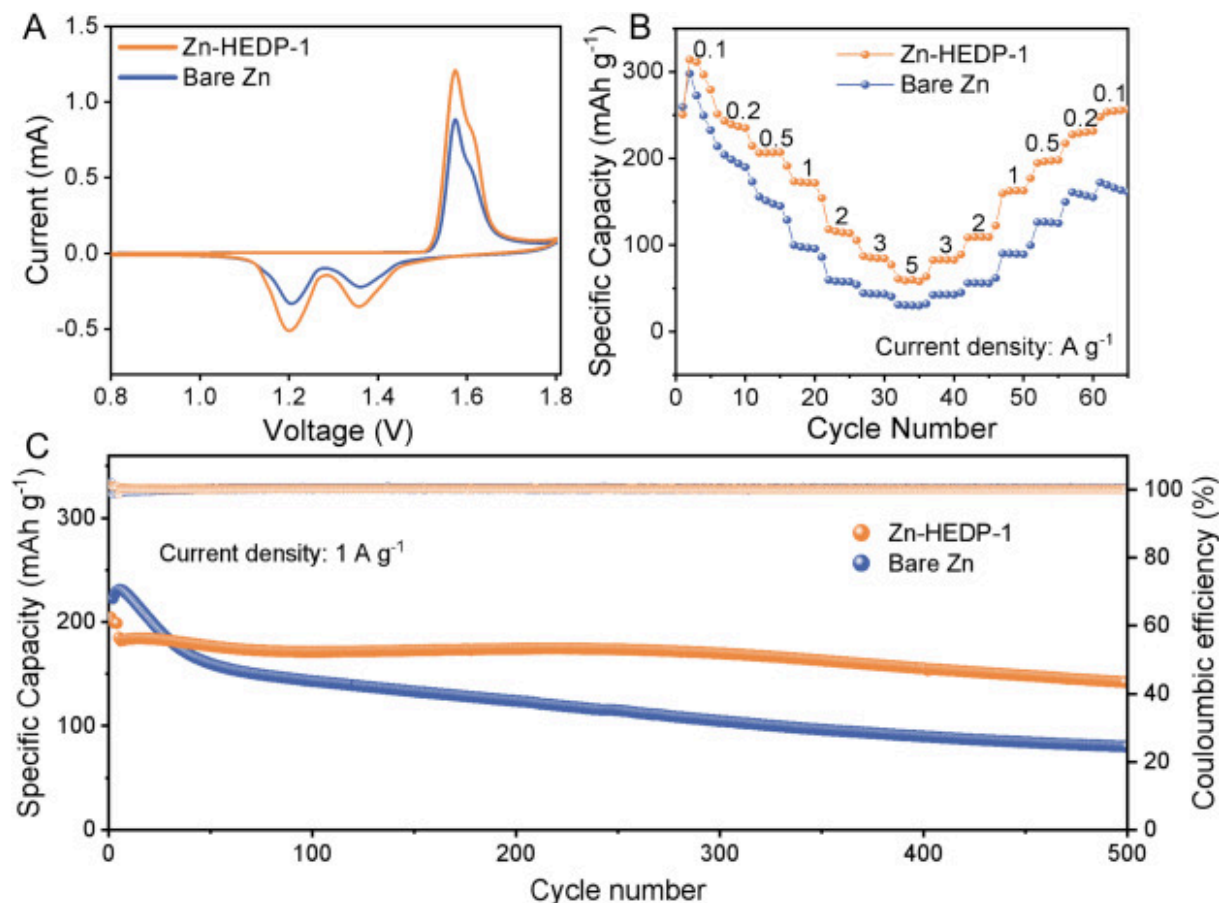
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Fig. 8. (A) CE curves at $2 \text{ mA cm}^{-2} / 1 \text{ mAh cm}^{-2}$, (B) Voltage profiles at 1 mA cm^{-2} , and (C) CE plots at $5 \text{ mA cm}^{-2} / 2.5 \text{ mAh cm}^{-2}$ of half cells with Zn-HEDP-1 and bare Zn electrodes.

The optimization of $\text{Zn}(\text{HEDP})_x$ interface on the deposition kinetics was quantitatively analyzed by evaluating the nucleation overpotential. Fig. 8B shows that the Zn-HEDP-1 electrode achieved a nucleation overpotential of only 23.3 mV under the condition of $1 \text{ mA cm}^{-2} / 1 \text{ mAh cm}^{-2}$, representing a 28.6% reduction compared to 32.5 mV of bare Zn. This phenomenon is attributed to the strong coordination between phosphonate groups in HEDP and Zn metal [43], which lowers the activation energy barrier for zinc deposition. It is noteworthy that the Zn-HEDP-1 half-cell maintains a stable operation for 730 cycles with an average CE of 99.84% even under more extreme conditions (5 mA cm^{-2} , 2.5 mAh cm^{-2}). In contrast, the bare Zn electrode exhibits a rapid performance degradation after only 141 cycles (Fig. 8C).

The Zn-HEDP-1//MnO₂ full cell was assembled to evaluate the feasibility of Zn-HEDP-1 anode in practical application. CV tests are firstly utilized to analyze the redox characteristics (Fig. 9A) and the results indicate that full cells with bare Zn and Zn-HEDP-1 exhibit two pairs of redox peaks, suggesting that the kinetic process of reversible reactions of Zn²⁺ and H⁺ intercalation/deintercalation remains highly consistent [44]. This confirms that the HEDP-modified Zn anode does not alter the intrinsic electrochemical behavior. In the rate-capability assessment, the Zn-HEDP-1//MnO₂ full cell demonstrates a superior performance (Fig. 9B), exhibiting a gradient response in the discharge capacity across a current density range of 0.1–5 A g⁻¹. At each current density, the Zn-HEDP-1 full cell delivers higher capacities compared to those of bare Zn. Long-term cycling stability experiments further reveal (Fig. 10C) that the Zn-HEDP-1 anode maintains a discharge capacity of 142.1 mAh g⁻¹ after 500 cycles at 1 A g⁻¹, achieving a high capacity retention rate of 69.6%. However, the full cell with bare Zn anode exhibits a discharge capacity of only 80.1 mAh g⁻¹ with a retention rate of 36.1%. The full-cell performance of this work is compared with recent studies in **Table S1**, in the supporting information. Considering the facile synthesis of Zn-HEDP, this work demonstrates the engineering application potential of HEDP molecules on interface regulation in rechargeable AZIBs.



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Fig. 9. (A) CV curves at 1 mV s⁻¹, (B) Rate-capability performance, and (C) Long-term cycling performance at 1 A g⁻¹ of full cells with Zn-HEDP-1 and bare Zn anodes.

4. Conclusions

In summary, the hydrophilic Zn(HEDP)_x layer effectively promotes the preferential deposition along the (002) planes, suppresses the zinc dendrite formation, and thereby enhances the long-term cycling life of electrode. In-situ characterizations confirm the uniform Zn deposition at the interface, thereby ensuring a stable electrochemical behavior. The symmetric cell with Zn-HEDP-1

achieves a stable cycling for over 2140 h at $1 \text{ mA cm}^{-2}/1 \text{ mAh cm}^{-2}$. The Cu//Zn-HEDP-1 half cell maintains an average Coulombic efficiency of 99.78% after 2100 cycles (2 mA cm^{-2} , 1 mAh cm^{-2}). The corresponding full cell retains a capacity retention rate of 69.6% after 500 cycles of charge-discharge at 1 A g^{-1} , significantly outperforming 36.1% of bare Zn. This work demonstrates the remarkable application potential of hydrophilic interface in advanced zinc-based energy storage devices.

CRedit authorship contribution statement

Zichuang Jiao: Writing – original draft, Investigation. **Hu Zhou:** Writing – review & editing. **Haopeng Chen:** Methodology, Investigation. **Mengfan Tang:** Validation. **Yanxin Qiao:** Resources. **Chunfeng Meng:** Writing – review & editing, Supervision. **Aihua Yuan:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We sincerely thank the financial support of National Natural Science Foundation of China (No. [22279047](#)).

Appendix A. Supplementary data

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Supplementary material

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Data availability

Data will be made available on request.

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